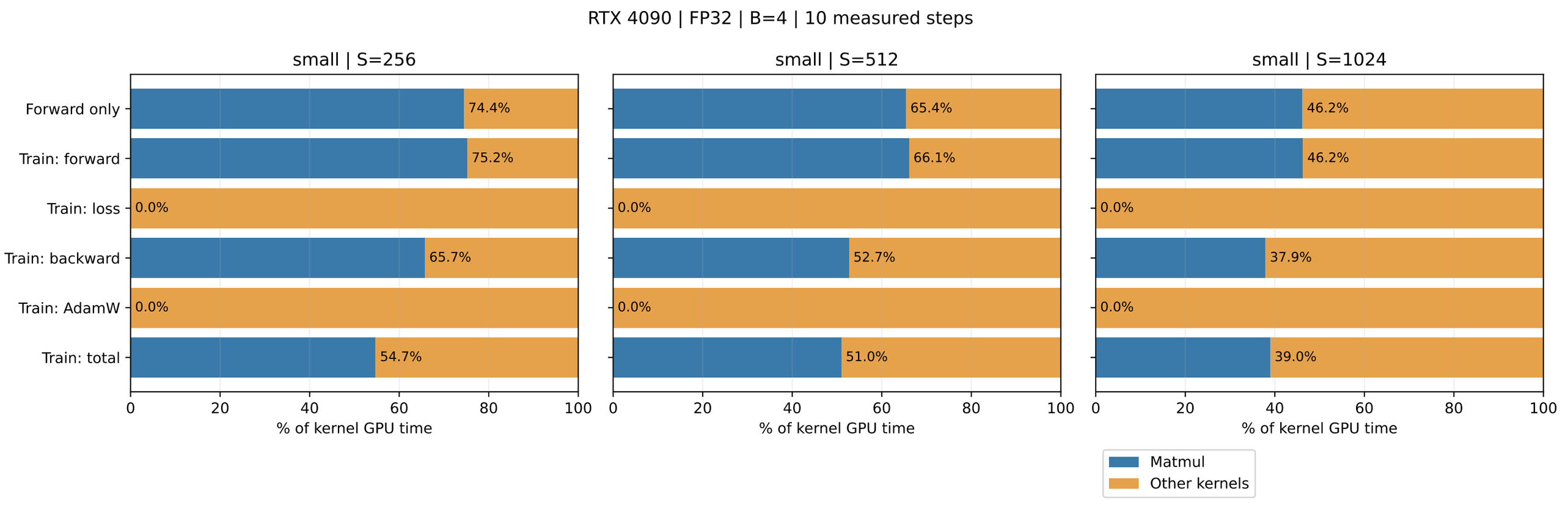
RTX 4090 | FP32 | B=4 | 10 measured steps
small | S=256
small | S=512
small | S=1024
Forward only
Train: forward
Train: loss
Train: backward
Train: AdamW
Train: total
74.4%
75.2%
0.0%
65.7%
0.0%
54.7%
65.4%
66.1%
0.0%
52.7%
0.0%
51.0%
46.2%
46.2%
0.0%
37.9%
0.0%
39.0%
0
20
40
60
80
100
% of kernel GPU time
Matmul
Other kernels